



Modelling Financial Markets

Chapter 0: Financial Markets in 2026



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AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material (1/2)

- ▣ AI: Artificial Intelligence
- ▣ AMM: Automated Market Maker
- ▣ AP: Authorised Participant
- ▣ AR: AutoRegressive (model)
- ▣ ASE: Academia de Studii Economice din București (the Bucharest University of Economic Studies)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BET-FI: BET Financial Investment companies index
- ▣ BET-TR: BET Total Return (index)
- ▣ BIS: Bank for International Settlements
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CAGR: Compound Annual Growth Rate
- ▣ CFTC: Commodity Futures Trading Commission
- ▣ COVID-19: COronaVirus Disease 2019
- ▣ CRIX: CRyptO IndeX
- ▣ EBA: European Banking Authority
- ▣ EMH: Efficient Market Hypothesis
- ▣ ESMA: European Securities and Markets Authority
- ▣ ETF: Exchange-Traded Fund
- ▣ ETP: Exchange-Traded Product
- ▣ EU: European Union
- ▣ FRED: Federal Reserve Economic Data
- ▣ FTSE: Financial Times Stock Exchange (FTSE Russell, index provider)
- ▣ FX: Foreign eXchange
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GENIUS: Guiding and Establishing National Innovation for U.S. Stablecoins (Act)

Acronyms used in this material (2/2)

- HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- HFT: High-Frequency Trading
- IPO: Initial Public Offering
- LLM: Large Language Model
- MDD: Maximum DrawDown
- MDE: Minimum Detectable Effect
- NAV: Net Asset Value
- NYSE: New York Stock Exchange
- OMV: Österreichische Mineralölverwaltung (Austrian Mineral Oil Administration (OMV group))
- OTC: Over-The-Counter
- ROL: Romanian old leu (ISO 4217 code, until 2005)
- RWA: Real-World Assets
- SE: Standard Error
- SEC: Securities and Exchange Commission
- SPDR: Standard & Poor's Depositary Receipts
- SR: Sharpe Ratio
- UDR: Uzinele și Domeniile Reșița (the Reșița Works and Domains)
- US: United States
- UTC: Coordinated Universal Time (Temps Universel Coordonné)
- VIX: CBOE Volatility Index

Outline

- Course Organisation
- Key Measures and Their Precision
- The Financial System
- Market Structure
- The 2000–2026 Landscape
- Passive Investing and Concentration
- Digital Assets and Tokenisation
- The Romanian Market
- Data and Tools
- AI for Scientific Discovery
- Conclusions
- References

About the Course

Programme

- Master's programme **Applied Statistics and Data Science**
- Faculty of Cybernetics, Statistics and Economic Informatics, Bucharest University of Economic Studies (ASE)
- 2 hours lecture + 2 hours seminar per week, academic year 2026/2027

Team

- Lecture: Prof. Daniel Traian Pele, danpele@ase.ro
- Seminar: Antoaneta Amza, PhD candidate, antoaneta.amza@csie.ase.ro
- Office hours: by appointment

What You Will Be Able to Do

- Describe how modern markets work
 - ▶ from exchanges and ETFs to tokenised assets and cryptocurrencies
- Build, estimate and test statistical models on real data
 - ▶ returns, volatility, risk, dependence and bubbles
- Evaluate models honestly
 - ▶ out-of-sample tests, backtests and proper scoring of forecasts
- Produce reproducible research in Python and GitHub
 - ▶ every chart documented as a Quantlet: code + description of inputs and outputs

Course Map: 20 Chapters

- ▣ 0 Financial Markets in 2026
- ▣ 1 Stylized Facts and Returns
- ▣ 2 Market Efficiency: From EMH to Adaptive Markets
- ▣ 3 Factor Models and Asset Pricing
- ▣ 4 Portfolio Optimization
- ▣ 5 Conditional Volatility: GARCH Models
- ▣ 6 Multivariate Volatility and Dependence
- ▣ 7 Value-at-Risk and Expected Shortfall
- ▣ 8 Backtesting and Evaluating Risk Forecasts
- ▣ 9 Realized Volatility
- ▣ 10 Market Microstructure
- ▣ 11 Continuous-Time Models
- ▣ 12 Options and the Volatility Surface
- ▣ 13 Machine Learning in Finance
- ▣ 14 Deep Learning and Time-Series Foundation Models
- ▣ 15 LLMs and Sentiment Analysis
- ▣ 16 Digital Assets and DeFi
- ▣ 17 Bubbles and Crashes
- ▣ 18 Systemic Risk and Networks
- ▣ 19 Review and Project Presentations

Assessment

Grading

- ▣ Team project (GitHub, presentation, oral defence): **70%**
- ▣ Quizzes and seminar activity: **20%**
- ▣ Attendance: **10%**
 - ▶ Required: at least **4 lectures** and at least **4 seminars**
- ▣ No written exam

Quizzes

- ▣ One quiz per chapter, on the course website
- ▣ 20 questions drawn at random from a larger bank
- ▣ Each answer is explained immediately

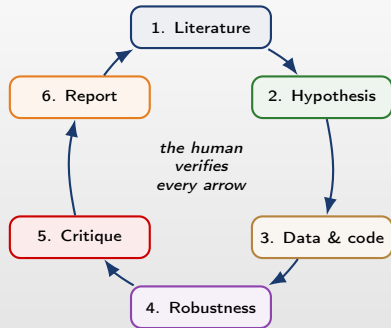
Attendance

- Minimum attendance: at least **4 lectures** and at least **4 seminars**
- Attendance is recorded in every lecture and seminar with a Google Form
 - ▶ The link is given in every lecture and seminar
 - ▶ Log in with your ASE account; name, group, session, chapter
- Each session has a short code written on the board
 - ▶ Only answers with the correct code, sent on the day of the session, count
 - ▶ The code changes every session and is given only in the room

The Team Project

- Teams of 3–4 students, one research question on real market data
- Public GitHub repository
 - ▶ code, a description of the data, results and a README file explaining how to run everything
- Every figure published as a Quantlet
 - ▶ one folder per result: code + `Metainfo.txt` (name, description, keywords, inputs, outputs) + chart
- Weekly commits: the history of the repository shows how the work progressed
- Final presentation in Chapter 19

The AI-Assisted Research Loop



- An AI assistant (an LLM such as Claude, ChatGPT or Gemini; GitHub Copilot in the editor) can speed up every step of empirical research
- Literature search: Semantic Scholar, Elicit, Google Scholar
- The loop is iterative: a failed robustness check or a strong critique sends you back to the hypothesis
- The human owns the question, the identification, the inference and every reported number

What AI Is Good At, What Stays Human

Delegate, then verify

- ▣ Mapping a literature: candidate papers, competing explanations, known pitfalls
- ▣ Enumerating the testable implications of each hypothesis and alternative specifications
- ▣ First drafts of estimation code, unit tests and plots
- ▣ Adversarial review: threats to the identification of your own design

- ▣ Typical LLM failures: invented references, plausible but wrong formulas, silent look-ahead in code, and a “significant” result found by searching until something passes

Stays human

- ▣ Choosing the question and the identifying assumption
- ▣ Fixing H_0 , the test, the sample and the multiple-testing correction before seeing results
- ▣ Checking every reference (DOI resolves, title matches) and every number against the data
- ▣ Judging economic versus statistical significance, and signing the result

How Every Lecture Ends

- ▣ Final section of every chapter: **AI for Scientific Discovery**
 - ▶ **An open question** from the chapter, with recent peer-reviewed references
 - ▶ **The discovery loop, step by step**: one researcher-level prompt per step
 - ▶ **What the researcher must check**: the chapter's failure points and the test that catches each
 - ▶ **A mini-case**: one pass of the loop on course data, from hypothesis to test to number
 - ▶ **Your turn**: a project seed, with a published number to replicate first
- ▣ Project rule: every AI use goes into [AI_USE.md](#), every caught AI error into [AI_ERRORS.md](#)
- ▣ Seminars practise the reverse skill: critique an AI answer (exercise C2)

Tools

Software

- ▣ Python with pandas, numpy, scipy, statsmodels, scikit-learn, matplotlib
- ▣ Jupyter notebooks, runnable in Google Colab without installation
- ▣ Git and GitHub for code, data and version history

Platforms

- ▣ **Quantlet**: a repository of documented code, one folder per chart
- ▣ **Quantinar**: video courses for advanced topics
- ▣ Course repository: github.com/danpele/MFM

Reproducibility Is Part of the Result

- Rule of the course: a result exists only if someone else can reproduce it from your repository
- In practice
 - ▶ the data are loaded by code, never copied manually
 - ▶ every number in a slide or report comes from a script
 - ▶ random numbers use a fixed seed, so reruns give the same result
- All charts of this lecture follow the rule
 - ▶ each chart has a Quantlet link at the bottom right of its slide

Core Reading

- ▣ Franke, Härdle & Hafner (2019), *Statistics of Financial Markets*, 5th ed.: the main statistical reference
- ▣ Campbell, Lo & MacKinlay (1997), *The Econometrics of Financial Markets*
- ▣ Brandimarte (2018), *An Introduction to Financial Markets: A Quantitative Approach*
- ▣ Härdle & Simar (2019), *Applied Multivariate Statistical Analysis*
- ▣ Härdle, Chen & Overbeck (2017) (Eds.), *Applied Quantitative Finance*, 3rd ed.
- ▣ Härdle, Harvey & Reule (2020), "Understanding cryptocurrencies", *Journal of Financial Econometrics*
 - ▶ Further papers are cited in each chapter; every citation is a clickable link

Prerequisites Recap (1/2): Prices, Returns, Growth

- ▣ **Price index** (BET, S&P 500) ignores dividends; **total-return index** (BET-TR) reinvests them; **adjusted close** does the same for a stock or ETF
- ▣ Simple return $R_t = P_t/P_{t-1} - 1$; log return $r_t = \ln(P_t/P_{t-1}) = \ln(1 + R_t)$, additive over time
- ▣ **CAGR** $= (P_{\text{end}}/P_{\text{start}})^{1/Y} - 1 = e^{\mu_{\log}} - 1$
 - ▶ Y = calendar days/365.25; $\mu_{\log}$ = mean log return per year; "x3.7" in a chart $= P_{\text{end}}/P_{\text{start}}$
 - ▶ S&P 500, 2 Jan 2015 – 18 Sep 2026: x3.72, CAGR 11.9%; with dividends (SPY) 13.7%
- ▣ **Annualised volatility** $\hat{\sigma}_d \sqrt{q}$, $\hat{\sigma}_d$ = standard deviation of daily log returns, q = observed trading days per year (252 equities, 365 crypto)
 - ▶ valid only for uncorrelated returns with constant variance (two slides ahead)

Prerequisites Recap (2/2): Drawdown, Correlation, Rates

- ▣ **Drawdown** $DD_t = P_t / \max_{s \leq t} P_s - 1$; **maximum drawdown** $= \min_t DD_t$
 - ▶ S&P 500: 1 565.15 (9 Oct 2007) to 676.53 (9 Mar 2009), -56.8% ; the peak was regained on 28 Mar 2013
- ▣ **Correlation**: Pearson correlation of returns (never of prices); **rolling**: on a moving 252-day window
 - ▶ two calendars (stocks vs crypto): join the prices on common days first, then compute returns
- ▣ **Yield to maturity**: the internal rate of return of the promised cash flows of a bond
 - ▶ equals the realised return only without default and with coupons reinvested at the same yield
- ▣ **Spread** $y^{10y} - y^{2y}$; **inversion**: spread < 0 ; units: percentage points (pp), 1 bp = 0.01 pp

How Precisely Do We Know an Expected Return?

- Daily returns iid with mean μ_d and standard deviation σ_d ; $T = qY$ observations over Y years
- Annualised mean $\hat{\mu} = q\bar{R}$: $SE(\hat{\mu}) = q\sigma_d/\sqrt{T} = \sigma_a/\sqrt{Y}$, with $\sigma_a = \sigma_d\sqrt{q}$
 - ▶ it depends only on the **span** Y , not on the sampling frequency ([Merton \(1980\)](#))
- S&P 500 with dividends (SPY), 2015–2026: $\hat{\mu} = 14.4\%$, $\sigma_a = 17.5\%$, $Y = 11.71$
 - ▶ $SE = 17.5\%/\sqrt{11.71} = 5.1$ pp; 95% confidence interval [4.3%, 24.4%]
 - ▶ halving the SE needs four times the span (47 years), not more frequent data
- Variance behaves differently: $Var(\hat{\sigma}^2) \approx \sigma^4(\kappa - 1)/T$ shrinks with T , slowly when the kurtosis κ is large (SPY daily: $\kappa = 17.1$, Chapter 1)

$\sqrt{h}$ Scaling and the Variance Ratio

- Stationary returns with variance σ^2 and autocorrelations ρ_j :

$$\text{Var}\left(\sum_{j=1}^h r_{t+j}\right) = h\sigma^2\left[1 + 2\sum_{j=1}^{h-1}\left(1 - \frac{j}{h}\right)\rho_j\right]$$

- The $\sqrt{h}$ rule is exact when returns are **uncorrelated** with **constant variance**; independence is not needed
- Variance ratio $\text{VR}(h) = \text{Var}(h\text{-day return})/(h\sigma^2)$; for an AR(1), $\text{VR}(h) \rightarrow (1 + \rho)/(1 - \rho)$
 - ▶ EUR/RON, BNR fixing: $\hat{\rho}_1 = 0.17$, factor $\sqrt{1.17/0.83} = 1.19$: annual volatility 4.4% $\rightarrow$ 5.3%
 - ▶ S&P 500, 2000–2026: $\hat{\rho}_1 = -0.10$, $\widehat{\text{VR}}(20) = 0.75$: $\sqrt{252}$ overstates the monthly risk
 - ▶ EUR/RON market file with bad ticks: $\hat{\rho}_1 = -0.57$, factor 0.52: $\sqrt{252}$ roughly doubles the risk
- Formal variance-ratio tests: Chapter 2

The Sharpe Ratio Is an Estimate

- ▣ Sharpe ratio (risk-free rate set to 0 here): $SR_d = \mu_d / \sigma_d$ per day, annualised $SR_a = \sqrt{q} SR_d$
- ▣ Delta method, iid returns: $\sqrt{T} (\widehat{SR}_d - SR_d) \xrightarrow{d} N(0, 1 + SR_d^2/2)$ (Lo (2002))
 - ▶ annualised from daily data: $\text{Var}(\widehat{SR}_a) = q(1 + SR_d^2/2)/T = (1 + SR_a^2/(2q))/Y \approx 1/Y$
 - ▶ SPY, 2015–2026: $\widehat{SR}_a = 0.82$, $SE = 0.29$, 95% confidence interval [0.25, 1.39]
- ▣ Skewness and fat tails: Opdyke (2007) correction (SPY: 0.30); dependent returns: HAC or studentised bootstrap (Ledoit & Wolf (2008))
- ▣ Picking the best of many assets or strategies ex post inflates the SR: deflated SR (Bailey & López de Prado (2014), Chapter 13)

What Financial Markets Are and What They Do

- **Financial market:** a mechanism, physical or electronic, where buyers and sellers exchange claims on future cash flows at prices formed by supply and demand
 - ▶ a **claim** can be a share, a bond, a currency, a contract or a token; the **price** summarises the views and needs of all participants
- Functions
 - ▶ **capital allocation:** savings flow to firms and governments; **risk sharing:** risk moves to those willing to bear it
 - ▶ **liquidity:** assets convert to cash quickly; **price discovery:** prices aggregate dispersed information
 - ▶ **payments and settlement:** transfer of ownership and cash; **monitoring:** prices discipline managers and governments
- Statistics of financial markets studies how these prices move

Asset Classes

Class	Examples	Cash flow	Main risks
Equities	shares, equity ETFs	dividends, residual claim	market, idiosyncratic
Fixed income	government and corporate bonds	coupons, principal	interest rate, credit
Money market	T-bills, repo	short-term interest	liquidity
FX	EUR/USD, EUR/RON	interest differential	currency, policy
Commodities	gold, oil futures	none (roll yield)	supply, storage
Derivatives	futures, options, swaps	contingent payoffs	leverage, counterparty
Digital assets	Bitcoin, Ether, stablecoins	none / protocol-specific	volatility, regulation

- The course models all of them with the same statistical toolbox: returns, volatility, dependence and tails

Asset Classes: Terms Used in the Table

- ▣ **ETF**: exchange-traded fund, a fund whose shares trade on an exchange like a stock
- ▣ **T-bills**: Treasury bills, government debt with maturity below one year
- ▣ **Repo**: repurchase agreement, a short-term loan backed by securities
- ▣ **FX**: foreign exchange, the currency market; EUR/RON is the price of one euro in lei
- ▣ **Futures**: a contract to buy or sell later at a price fixed today; **roll yield** is the gain or loss from replacing an expiring contract with the next one
- ▣ **Options**: the right, not the obligation, to buy or sell at a fixed price; **swaps**: exchanges of cash-flow streams
- ▣ **Idiosyncratic risk**: specific to one company; **counterparty risk**: the other side of a contract may not pay

Market Participants

- **Buy side:** invests money and buys market services
 - ▶ retail investors; institutions (pension funds, insurers, sovereign funds); asset managers, **passive** (replicate an index) or **active** (try to beat it); hedge funds and proprietary traders, often leveraged
- **Sell side:** provides trading services
 - ▶ **brokers** execute client orders; **dealers** trade from inventory; **market makers** quote a buy and a sell price and earn the difference
- **Infrastructure**
 - ▶ exchanges and **multilateral trading facilities (MTFs)**; **central counterparties (CCPs)** become the buyer to every seller and the seller to every buyer; **custodians** keep the securities
- Different horizons and information sets are why prices move in ways that simple models miss

Where Trading Happens

Exchanges

- Central limit order book
- Quotes and trades are public
- Examples: NYSE, Nasdaq, BVB

OTC markets

- Bilateral trades with dealers
- Dominant for bonds, FX, swaps
- Prices less visible

Dark pools

- No quotes shown before the trade
- Large orders hide their size
- Prices referenced to public markets

- **OTC**: over the counter, trades negotiated directly between two parties, without an exchange
- NYSE: New York Stock Exchange; BVB: Bursa de Valori București (Bucharest Stock Exchange)

Crypto Trading Venues

- **Centralised exchanges:** companies that run an order book, like a stock exchange, but open 24 hours a day, 7 days a week
- **Decentralised exchanges:** programs on a blockchain (smart contracts) that trade without an operator
 - ▶ most use an **automated market maker (AMM)**: a pool of two tokens whose price is set by a formula (Chapter 16)
- The same coin trades on hundreds of venues at slightly different prices
- **Makarov & Schoar (2020)**: large and persistent price differences across exchanges and countries
 - ▶ fragmentation creates arbitrage opportunities that capital controls and costs keep open

The Limit Order Book



- **Limit order book:** the list of all waiting buy and sell orders, sorted by price
- **Bid:** an offer to buy; the best bid is the highest one (99.9 here)
- **Ask:** an offer to sell; the best ask is the lowest one (100.01 here)
- **Bid-ask spread:** best ask minus best bid, the price of trading immediately
- **Depth:** the quantity waiting near the best prices
 - ▶ Illustration only; Chapter 10 models the book with real data

Order Types and Market Impact

- **Limit order**: at a given price or better; waits in the book and **provides** liquidity
- **Market order**: now, at the best available prices; **consumes** liquidity
- **Stop order**: becomes a market order at a trigger price; clustered stops can accelerate a fall
- **Liquidity**: trading a large quantity quickly without moving the price much
- Example, book of the previous slide: a market buy of 5 units
 - ▶ 2 units at 100.01 and 3 units at 100.02: average $(2 \times 100.01 + 3 \times 100.02)/5 = 100.016$
 - ▶ the best ask moves to 100.02 and the spread widens from 0.11 to 0.12: **market impact** (Chapter 10)

Auctions versus Continuous Trading

Call auctions

- ▣ Orders are collected, then matched at one price that maximises traded volume
- ▣ Used at the open and close of most exchanges
- ▣ Concentrate liquidity and reduce noise in reference prices

Continuous trading

- ▣ Every incoming order is matched immediately if possible
- ▣ Priority: best price first, then the earliest order
- ▣ Speed matters: being first in the queue is rewarded

Why Market Structure Matters for Statistics

- For exchange-traded assets, the “daily close” used in this course is the price of the closing auction
 - ▶ exceptions: Bitcoin trades 24/7 with no auction, so its daily price is a snapshot at 00:00 UTC; EUR/RON is the BNR fixing, published around 13:00 Bucharest time (Chapter 1)
- **High-frequency trading (HFT)**: computer programs that trade in microseconds
 - ▶ Budish, Cramton & Shim (2015) argue that continuous trading creates a speed arms race and propose frequent batch auctions instead
- Bid–ask bounce makes observed price changes negatively autocorrelated: $\text{Cov}(\Delta p_t, \Delta p_{t-1}) = -s^2/4$
 - ▶ so the spread can be estimated from prices alone, $\hat{s} = 2\sqrt{-\hat{\gamma}_1}$ (Roll (1984); Chapter 10)
- Spreads, impact and fragmentation add noise to observed prices, most visible at high frequency (Chapter 9)

Recap: Market Structure

- Trading happens on exchanges, OTC with dealers, in dark pools and, for crypto, on centralised and decentralised venues
- The limit order book stores waiting orders; market orders consume them
- The spread and the depth measure how costly it is to trade
- Auctions set the reference prices; continuous trading rewards speed

From the Trading Floor to Screens



NYSE trading floor, 2011

Photo: Carol M. Highsmith (2011);
public domain; Wikimedia Commons



Electronic displays, Chicago Board of Trade, 1992

Photo: Lars Plougmann (1992); CC BY-SA 2.0;
Wikimedia Commons

- **Open outcry:** traders met on the floor and matched orders by voice and hand signals
- Electronic limit order books gradually replaced most floor trading
- Today most volume is matched by computers (market microstructure, Chapter 10)

Crashes Before Our Data: Black Tuesday, 1929

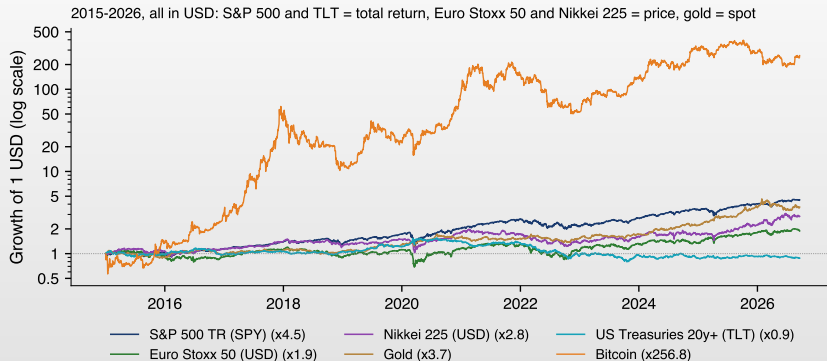


Crowd outside the NYSE, 29 October 1929

Photo: U.S. government (1929); public domain;
[Wikimedia Commons](#)

- 29 October 1929 (**Black Tuesday**): crowds gather outside the New York Stock Exchange during the crash
- The crash of October 1929 opened the Great Depression of 1929–1933
- Long, deep drawdowns are not new: the next slides measure the 2000–2026 episodes with the same tools
- Bubbles and crashes are modelled in Chapter 17

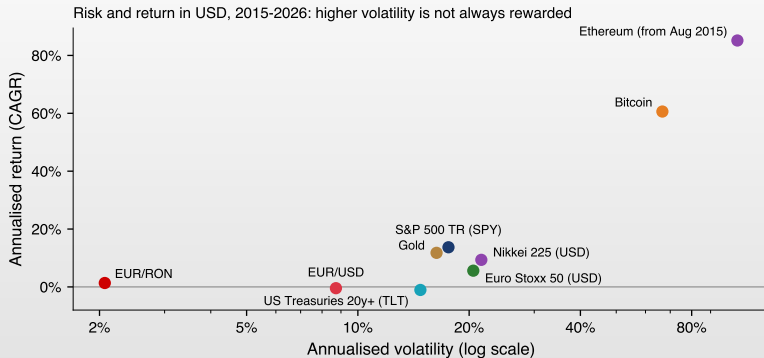
Asset Classes since 2015



Reading the Cross-Asset Chart

- Each line shows the value of 1 USD invested on 2 Jan 2015; every series is in USD, on a log scale
 - ▶ Euro Stoxx 50 and Nikkei 225 converted with daily EUR/USD and USD/JPY rates from [FRED](#)
- S&P 500 with dividends reinvested (SPY, total return) x4.5; gold x3.7; Nikkei 225 x2.8; Euro Stoxx 50 x1.9
- **TLT** (iShares 20+ Year Treasury Bond ETF, long US government bonds, total return) ended **below** its 2015 level: x0.9
- Bitcoin multiplied by about 257, from 315 USD to 80 901 USD
- Euro Stoxx 50 and Nikkei 225 are price indices: their dividends are missing
 - ▶ for the S&P 500 the gap is 1.8 pp a year: 13.7% with dividends (SPY) vs 11.9% for the price index

Risk and Return, 2015–2026



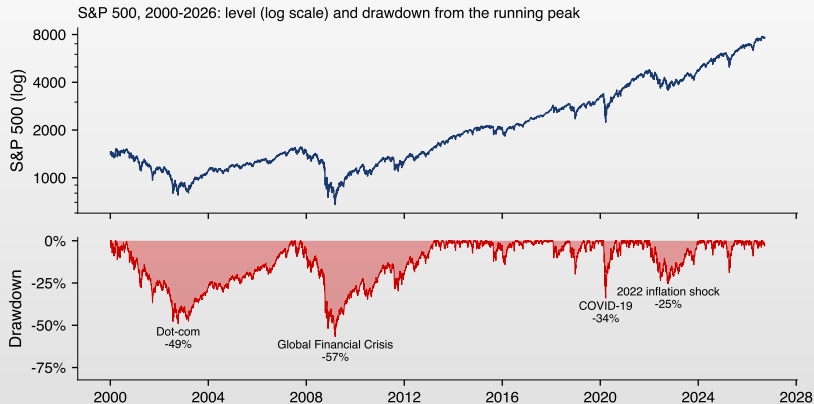
Reading the Risk–Return Chart

- Vertical axis: CAGR; horizontal axis: annualised volatility of daily log returns (both defined in Section 2)
- All in USD. S&P 500 with dividends (SPY): 13.7% a year, 17.6% volatility; Nikkei 225: 9.4%, 21.6%; Euro Stoxx 50: 5.6%, 20.5%
- Gold: 11.8% a year with 16.3% volatility, slightly below the S&P 500 (weekday quotes only; weekend prints removed)
- Bitcoin: 60.6% a year, 66.7% volatility; Ethereum (from Aug 2015): 85.2% and 106%
- EUR/RON: volatility 2.1% a year, the profile of a managed currency
- Ex post only: returns earned in the past are not forecasts

Is Any Ranking in the Risk–Return Chart Significant?

- ▣ Standard error of an annualised mean return, $\sigma_a/\sqrt{Y}$ over $Y = 11.7$ years
 - ▶ S&P 500 with dividends 5.1 pp, gold 4.8 pp, Euro Stoxx 50 6.0 pp, Nikkei 225 6.2 pp, Bitcoin 19.4 pp
- ▣ S&P 500 minus gold, common days: mean difference +1.9 pp a year, SE 6.8 pp, $t = 0.28$
- ▣ Sharpe ratios: S&P 500 0.82, gold 0.77, Nikkei 225 0.53, Euro Stoxx 50 0.37, TLT 0.00, Bitcoin 1.05
 - ▶ every SE $\approx 1/\sqrt{11.7} = 0.29$: a 95% interval is about ± 0.57
 - ▶ S&P 500 minus gold: +0.06, stationary-bootstrap 95% interval $[-0.65, +0.81]$ (Seminar 0)
- ▣ 11.7 years cannot rank the equity indices and gold; the chart describes the sample, it does not estimate expected returns precisely

Crises and Drawdowns

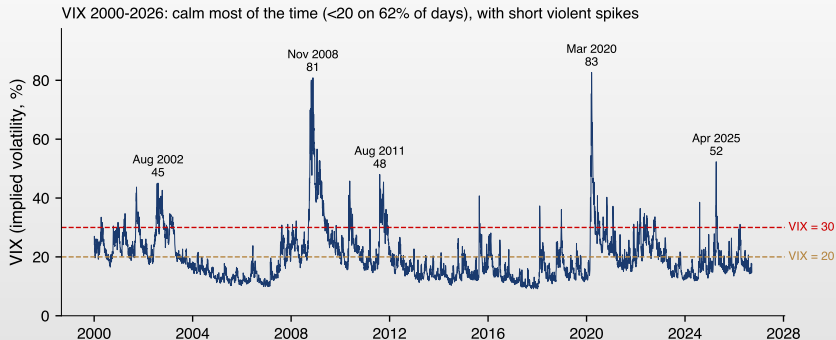


Reading the Drawdown Chart

- Lower panel: the drawdown $DD_t = P_t / \max_{s \leq t} P_s - 1$ of the S&P 500
- Four large troughs since 2000
 - ▶ −49% (Oct 2002): the end of the dot-com bubble
 - ▶ −57% (Mar 2009): the global financial crisis
 - ▶ −34% (Mar 2020): the COVID-19 crash, the fastest of the four
 - ▶ −25% (Oct 2022): the inflation and interest-rate shock
- Recovery can take years: after 2009 the index needed until March 2013 to regain its 2007 peak

 MFM_ch0_drawdowns_vix

Volatility Regimes: the VIX



MFM_ch0_drawdowns_vix

Reading the VIX Chart

- **VIX**: the CBOE Volatility Index, computed by the Chicago Board Options Exchange
 - ▶ the volatility of the S&P 500 over the next 30 days that is **implied** by option prices, in % per year
 - ▶ often called the “fear index”
- Since 2000 the *implied* volatility was below 20 on 62% of days and above 30 on 9%
- Largest peaks: 82.7 (16 Mar 2020), 80.9 (20 Nov 2008), 52.3 (8 Apr 2025)
- Calm periods are long, stress periods short and intense: volatility comes in regimes (Chapters 5 and 9)

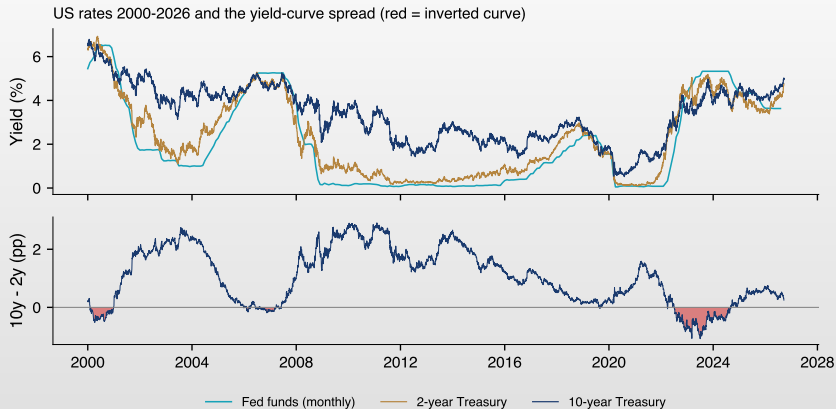
 MFM_ch0_drawdowns_vix

The VIX Is Not a Volatility Forecast

- $VIX^2 \approx$ the **risk-neutral** expectation of the S&P 500 variance over the next 30 days, computed from option prices
- Under the real-world probability, implied variance exceeds realised variance on average: the **variance risk premium** (Carr & Wu (2009))
 - ▶ investors pay to be insured against volatility spikes
- S&P 500, 2000–2026, realised volatility over the next 21 trading days
 - ▶ mean VIX 19.8 vs mean realised 16.2; the VIX exceeded the subsequent realised volatility on 83% of days
- The premium, implied minus realised variance, predicts equity returns at a quarterly horizon (Bollerslev, Tauchen & Zhou (2009))
- Implied volatility is therefore a biased forecast of realised volatility (Chapters 9 and 12)

 MFM_ch0_drawdowns_vix

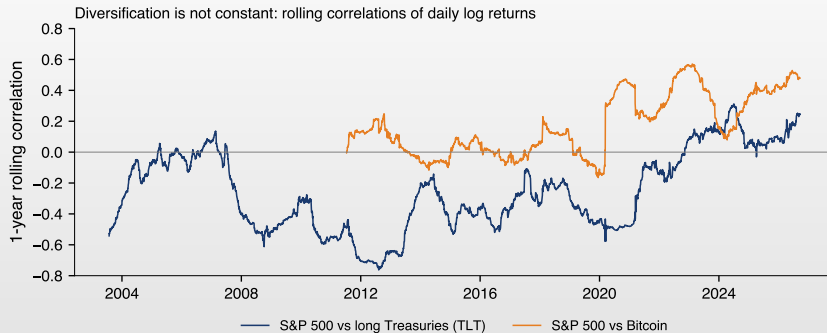
Interest Rates and the Yield Curve



Reading the Rates Chart

- Data: [FRED](#) (Federal Reserve Economic Data, St. Louis Fed)
 - ▶ 10-year and 2-year US Treasury yields; the **fed funds rate**, the overnight rate targeted by the US central bank
- The 10y – 2y spread first turned negative briefly on 1–4 Apr 2022 (2 trading days)
 - ▶ persistent inversion from 6 Jul 2022 to 26 Aug 2024: 537 trading days, minimum -1.08 pp (3 Jul 2023)
- 17 Sep 2026: 10-year 4.94%, 2-year 4.67%, spread $+0.27$ pp = 27 bp; fed funds 3.63% (Aug 2026)
- Rates set the discount factor of every asset in this course

Diversification Is Not Constant



Reading the Correlation Chart

- 1-year (252-day) rolling correlations of daily log returns; prices joined on common trading days first, then returns
- Stocks vs. long Treasuries (S&P 500 vs. TLT)
 - ▶ mean of the 252-day rolling correlation: -0.43 in 2010–2020, $+0.08$ in 2022–2026 (whole-period correlation: -0.47 , $+0.11$; Seminar 0)
 - ▶ after the 2022 inflation shock bonds hedged equities much less
- Stocks vs. Bitcoin
 - ▶ mean of the 252-day rolling correlation: 0.01 in 2016–2019, 0.38 in 2022–2026 (whole-period 2022–2026: 0.42)
 - ▶ crypto now moves more with equities than it did before 2020
- Portfolio models that assume a fixed correlation miss these shifts (Chapters 4 and 6)

Has the Stock–Bond Correlation Really Changed?

- ▣ Whole-period correlation S&P 500 – TLT: -0.47 in 2010–2020 ($n = 2\,769$), $+0.11$ in 2022–2026 ($n = 1\,182$)
- ▣ Fisher z test, iid pairs: $\operatorname{atanh} \hat{\rho}$ has variance $1/(n - 3)$
 - ▶ $z = 17.9$, $p < 10^{-10}$; iid 95% interval for the change $+0.58$: $[0.52, 0.65]$
- ▣ Volatility clustering makes pairs dependent: stationary bootstrap, mean block 20 days
 - ▶ 95% interval $[0.49, 0.68]$, 1.54 times wider; the change survives (Seminar 0, Ex3 Extended)
- ▣ The mean of 252-day rolling correlations and the whole-period correlation are different estimands

 MFM_ch0_correlations

Asynchronous Closes

- Bitcoin's daily price is a snapshot at 00:00 UTC; the S&P 500 closes at 20:00 or 21:00 UTC
 - ▶ news arriving between the two times enters the two series on different days
- With a common factor, the same-day correlation is attenuated and the rest appears in lead/lag correlations
 - ▶ correction: sum the lead, same-day and lag correlations (Scholes & Williams (1977); Lo & MacKinlay (1990)), or use longer returns (Burns, Engle & Mezrich (1998))
- Bitcoin – S&P 500, 2022–2026, daily on common days
 - ▶ same day 0.42; S&P 500 with Bitcoin one day earlier +0.03, one day later −0.05; sum 0.40
 - ▶ the asynchronous-close bias is small for this pair at the daily frequency

 MFM_ch0_correlations

Correlation in Turbulent Periods: the Volatility Bias

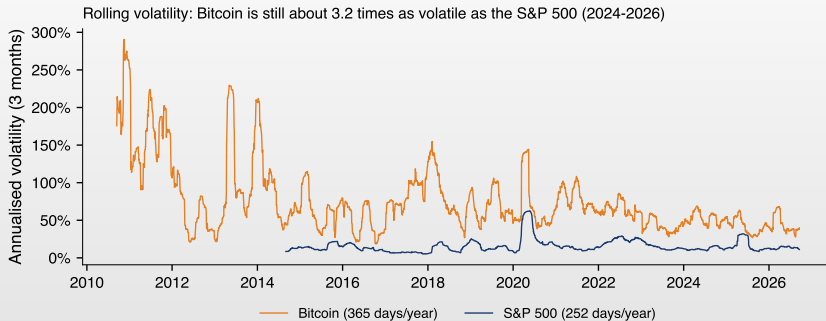
- A constant true correlation ρ , estimated only on days when the variance of the source market is $(1 + \delta)$ times higher, gives (Forbes & Rigobon (2002))

$$\rho^* = \rho \sqrt{\frac{1 + \delta}{1 + \delta \rho^2}} > |\rho| \text{ in absolute value}$$

- Bitcoin – S&P 500, 2022–2026, days split by the VIX (top quartile: $VIX > 21.8$)
 - ▶ 0.52 on high-VIX days vs 0.32 on the others; S&P 500 variance ratio $1 + \delta = 4.64$
 - ▶ adjusted $\hat{\rho} = \rho^* / \sqrt{1 + \delta(1 - \rho^{*2})} = 0.27$: the “stress correlation” is mostly a volatility effect
- The adjustment assumes no omitted common shock and no feedback between the two markets

 MFM_ch0_correlations

Volatility Differs by an Order of Magnitude



Reading the Rolling Volatility Chart

- Rolling 3-month (63 trading days) annualised volatility
 - ▶ Bitcoin annualised with $\sqrt{365}$, the S&P 500 with $\sqrt{252}$
- Bitcoin: average 65% in 2015–2017, 46% in 2024–2026
- S&P 500 in 2024–2026: 14%, so Bitcoin is still about **3.2 times** as volatile
- Volatility changes over time for both: a constant-volatility model is a poor description

 MFM_ch0_correlations

Selection, Survivorship and Horizon Effects

- ▣ **Survivorship:** indices and databases drop failed firms and coins, so past returns look too good (Brown, Goetzmann, Ibbotson & Ross (1992))
- ▣ **Selection:** Bitcoin's $\times 257$ is chosen ex post among thousands of coins; a best-of-many Sharpe ratio needs a deflated SR
- ▣ **Horizon:** the maximum drawdown grows with the sample length
 - ▶ driftless Brownian motion, log scale: $E[\text{MDD}] = \sqrt{\pi/2} \sigma \sqrt{T}$ (Magdon-Ismail et al. (2004))
 - ▶ $\sigma = 17.6\%$: about -50% over 10 years, -75% over 40 years
- ▣ The BET's -82.5% (29 years) and an MDD from 11.7 years of data are not comparable without the same T

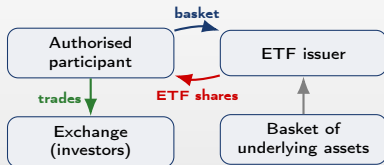
Recap: the 2000–2026 Landscape

- ▣ Returns differ widely across assets, but over 11.7 years most of the differences are not significant
- ▣ Deep drawdowns recur, recoveries can take years, and the maximum drawdown grows with the horizon
- ▣ Volatility comes in regimes; the VIX adds a variance risk premium on top of expected volatility
- ▣ Correlations change with the macro regime; tests must allow for dependence, asynchronous closes and volatility bias
- ▣ Crypto remains several times more volatile than equities, and more correlated with them than before

Exchange-Traded Funds

- ▣ An **ETF** is an investment fund whose shares trade on an exchange like a stock
- ▣ **Net asset value (NAV)**: the value of the fund's holdings per share, computed once a day
- ▣ Most ETFs are **passive**: they replicate an index at low cost
- ▣ Advantages: low fees, trading during the day, diversification in one security
 - ▶ in the US also tax efficiency: in-kind redemptions let the fund avoid distributing capital gains (a US tax rule, not a general feature)
- ▣ ETFs now package equities, bonds, commodities and, in the US since January 2024, spot Bitcoin

How ETF Creation and Redemption Work



- **Authorised participant (AP):** a large dealer allowed to deal directly with the issuer
- **Creation:** the AP delivers the basket of underlying assets and receives new ETF shares
- **Redemption:** the AP returns ETF shares and receives the basket
- This two-way channel keeps the market price close to the NAV

Worked Example: Arbitrage Keeps the ETF Price Near NAV

- ▣ Illustrative numbers: $NAV = 100$, the ETF trades at 100.50; **premium** $(P - NAV)/NAV = 0.5\%$
- ▣ The AP buys the basket for 100, delivers it to the issuer, receives one new ETF share and sells it for 100.50
 - ▶ 0.50 per share before costs; the extra ETF shares push the price down until the premium disappears
 - ▶ at a **discount** ($P < NAV$) the AP buys ETF shares and redeems them
- ▣ Statistical view: an AR(1) premium, $\pi_t = \phi \pi_{t-1} + u_t$, with half-life $\ln 0.5 / \ln \phi$
 - ▶ premiums are persistent where the NAV is stale or the basket is illiquid, e.g. foreign stocks or bonds (Petajisto (2017))

ETFs and Volatility

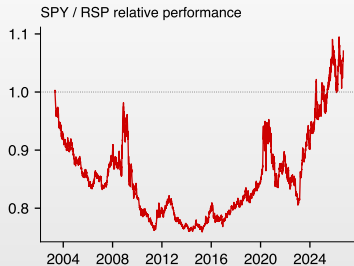
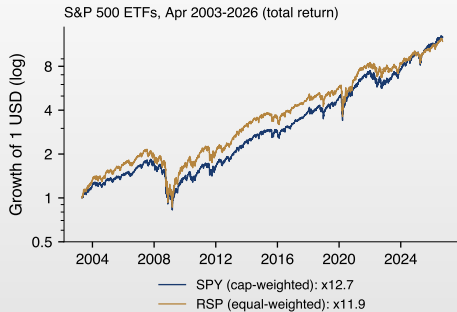
- ETF shares are traded more often and by faster investors than the underlying stocks
- Arbitrage transmits ETF price shocks to the underlying stocks
- Ben-David, Franzoni & Moussawi (2018): stocks with higher ETF ownership are more volatile
- For modelling: the growth of passive investing can change the statistical behaviour of the stocks it holds

Cap-Weighted versus Equal-Weighted Indices

Definition 1 (Index weights)

- Capitalisation weights: $w_i = \frac{MC_i}{\sum_{j=1}^n MC_j}$, with market capitalisation $MC_i = \text{price} \times \text{number of shares}$
 - Equal weights: $w_i = 1/n$ for every stock, rebalanced periodically
-
- SPY** (SPDR S&P 500 ETF): the S&P 500 with capitalisation weights
 - RSP** (Invesco S&P 500 Equal Weight ETF): the same 500 stocks, each with weight $1/500$
 - Ratio SPY/RSP rising: the largest companies are outperforming the average company

Market Concentration: SPY versus RSP



Reading the Concentration Chart

- Both ETFs are total return (adjusted close), since RSP's launch in April 2003
- Over the whole period the two grew similarly: SPY $\times 12.7$, RSP $\times 11.9$
- The SPY/RSP ratio rose by 31% since January 2023
- Over the whole sample the ratio ends at 1.07 times its April 2003 level
- Interpretation: a handful of very large companies drove the index in 2023–2026

 MFM_ch0_concentration

Why Concentration Matters for Modelling

For investors

- ▣ A cap-weighted index is less diversified than its number of stocks suggests
- ▣ Index risk depends on a few firms and one theme
- ▣ Passive flows buy the largest firms mechanically

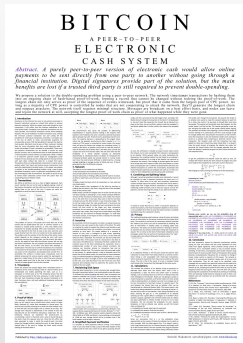
For statisticians

- ▣ Effective number of stocks $N_{\text{eff}} = 1 / \sum_i w_i^2$ (inverse Herfindahl): 500 for RSP, far fewer for a cap-weighted index (Chapter 4)
- ▣ Factor exposures shift over time (Chapter 3)
- ▣ Bubbles in a few names can dominate the index (Chapter 17)

Recap: Passive Investing

- ETFs give cheap, tradable access to whole indices and asset classes
- Creation and redemption by authorised participants keep ETF prices close to NAV
- Passive investing grows with capitalisation weights, which concentrates the index in the largest firms
- Concentration reduces diversification and changes the risk of the index

Where It Started: the Bitcoin White Paper



First page of the Bitcoin white paper (2008)

Satoshi Nakamoto (2008); CC0; Wikimedia Commons

- 31 October 2008: *Bitcoin: A Peer-to-Peer Electronic Cash System*, by the pseudonymous (real identity unknown) Satoshi Nakamoto [Nakamoto \(2008\)](#)
- A public ledger (the **blockchain**) kept by the network itself, without a trusted intermediary
- **Proof of work**: participants spend computing power to add blocks, which makes rewriting the history costly
- From this paper to spot ETFs, stablecoins and tokenisation: the next slides

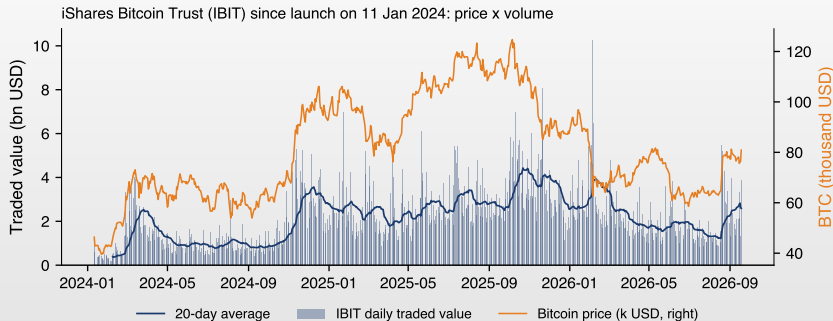
Cryptocurrencies: Basic Terms

- **Blockchain:** a shared ledger of transactions, copied on many computers and extended in blocks
- **Bitcoin** (2009): the first cryptocurrency, with a fixed supply rule (at most 21 million coins)
- **Ethereum:** a blockchain that runs programs, called **smart contracts**; its coin is Ether
- **Token:** a digital unit recorded on a blockchain, representing a coin, a claim or a right
- Trading is continuous, global and fragmented across many exchanges

Cryptocurrencies as an Asset Class

- ▣ Härdle, Harvey & Reule (2020): statistical properties, pricing and regulation of cryptocurrencies
- ▣ Pele et al. (2023): tail, memory and moment factors of returns set cryptos apart from stocks, FX, commodities, bonds and real estate
- ▣ Makarov & Schoar (2020): large and persistent price differences across exchanges and countries
- ▣ Our data: correlation with equities (mean of the 252-day rolling correlation) rose from about 0 (2016–2019) to 0.38 (2022–2026)
- ▣ Chapter 16 covers crypto indices (CRIX, the CRYPTO Index), stablecoins, AMMs and on-chain data in depth

Spot Bitcoin ETFs



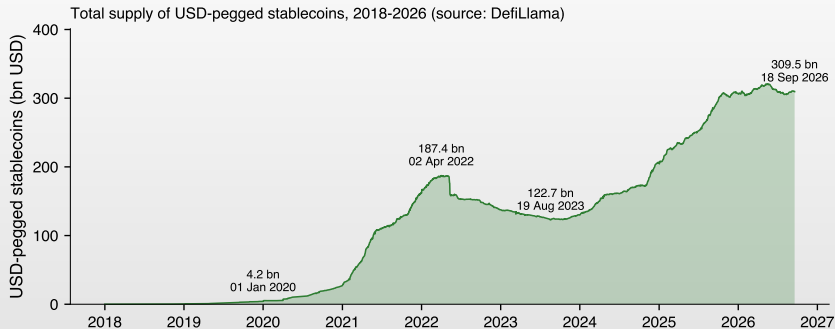
Reading the Spot Bitcoin ETF Chart

- **Spot** ETF: holds the asset itself (bitcoins), not futures contracts on it
- **ETP**: exchange-traded product, the general category that includes ETFs
- **SEC (2024)**: approval of spot Bitcoin ETPs in the United States, 10 January 2024
- **IBIT** (iShares Bitcoin Trust), trading since 11 Jan 2024
 - ▶ daily traded value = price $\times$ number of shares traded
 - ▶ median 1.97 bn USD per day, peak 10.3 bn USD (5 Feb 2026)

What Is a Stablecoin Peg?

- **Stablecoin:** a token designed to keep a fixed value, usually 1 USD; usually, but not always, backed by reserves
- **Peg:** the target exchange rate, e.g. 1 token = 1 USD
 - ▶ **reserve-backed:** the issuer holds cash and short-term government bonds and redeems tokens at 1 USD
 - ▶ **algorithmic:** the peg relies on a second token and trading rules, without full reserves
- **Depeg:** the market price moves away from the peg
- Our data, Tether (USDT), 2019–2026: within $\pm 0.5\%$ of 1 USD on 93% of days and within $\pm 1\%$ on 98%
 - ▶ extremes 0.974 and 1.054 USD, both in the COVID-19 panic of March 2020
- The algorithmic stablecoin TerraUSD lost its peg and collapsed in May 2022

Stablecoin Supply



Reading the Stablecoin Chart

- **Supply:** the total value of USD-pegged stablecoins in circulation, from [DefiLlama](#), a public DeFi data platform
- 4.2 bn USD (1 Jan 2020), peak 187.4 bn (2 Apr 2022), trough 122.7 bn (19 Aug 2023), 309.5 bn (18 Sep 2026)
- The 2022–2023 drop followed the collapse of TerraUSD and the crypto crash of 2022
- Regulated in the European Union by [MiCA, Regulation \(EU\) 2023/1114](#) (Markets in Crypto-Assets) and in the US by the [GENIUS Act, Public Law 119-27](#)

Tokenisation of Real-World Assets

Idea

- **RWA**: real-world assets, such as bonds, fund shares, deposits or commodities
- A token on a shared ledger represents a claim on such an asset
- Settlement and transfer become programmable and near-instant
- Examples: tokenised money-market funds and Treasury funds

- Statistical question: do tokenised assets inherit the properties of the underlying, or those of crypto markets?

Central-bank view

- **BIS**: Bank for International Settlements, the bank of central banks
- **BIS (2023)**: a “unified ledger” combining central bank money, deposits and assets
- **BIS (2025)**: stablecoins fall short of the three tests for money
 - ▶ singleness (accepted at par by all), elasticity (supply follows payment needs), integrity (resists financial crime)

Regulation of Digital Assets

	European Union	United States
Framework	MiCA, Regulation (EU) 2023/1114	GENIUS Act, Public Law 119-27 for payment stablecoins
Crypto ETPs	exchange-traded notes on crypto	spot Bitcoin ETPs approved in January 2024 (SEC (2024))
Supervisors	ESMA, EBA, national authorities	SEC, CFTC, banking regulators

- ESMA: European Securities and Markets Authority; EBA: European Banking Authority
- SEC: Securities and Exchange Commission; CFTC: Commodity Futures Trading Commission
- For modelling: regulatory dates are natural break points and event-study dates

Recap: Digital Assets

- ▣ Cryptocurrencies are a statistically distinct asset class, mainly through their heavy tails
- ▣ Spot ETFs and stablecoins connected crypto to mainstream finance
- ▣ Stablecoins hold their peg most of the time, but pegs can break
- ▣ Tokenisation brings traditional assets onto blockchains; regulation (MiCA, GENIUS Act) shapes who can hold them

The Bucharest Stock Exchange, 1882–1948

- **1 Dec 1882:** the securities and commodities exchange opens under a royal decree of King Carol I [BVB \(history\)](#)
- Early shocks [BVB press release \(2017\)](#)
 - ▶ the 1907 peasants' revolt and the Balkan war of 1912–1913
 - ▶ closed during the First World War
- After reopening: about 7 years of strong growth, then about 7 years of accelerated decline, overlapping the Great Depression of 1929–1933 [BVB press release \(2017\)](#)
- **1929:** a new law on exchanges, with the explanatory memorandum of minister Virgil Madgearu [Lege asupra burselor \(1929\)](#)
- **1948:** trading stops with the communist regime and the nationalisation of the economy [BVB \(history\)](#)



Share of UDR (Uzinele și Domeniile Reșița), June 1926. Scan: [EDHAC e.v.](#); public domain; [Wikimedia Commons](#)

The Palatul Bursei: Home of the Exchange



Photo: Nicolae Ionescu (1928); public domain; Wikimedia Commons



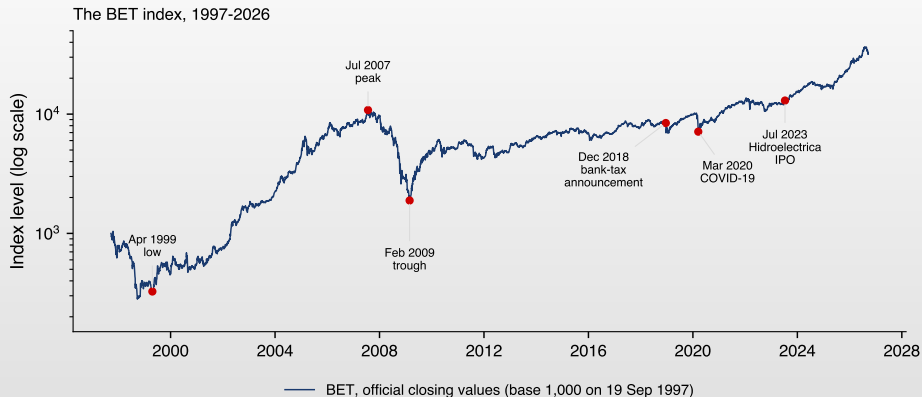
Photo: Chainwit. (2023); CC BY 4.0; Wikimedia Commons

- Built 1906–1912, architect Ștefan Burcuș, Str. Ion Ghica 4 [Palatul Bursei](#)
- Seat of the Exchange and of the Bucharest Chamber of Commerce and Industry
- From 1955: the State Central Library (today the National Library)
- Left: stockbrokers at the entrance, March 1928; right: the building in 2023

Interruption and Rebirth, 1948–2023

- ▣ **1948–1995**: almost 50 years without a stock exchange [BVB press release \(2017\)](#)
- ▣ **20 Nov 1995**: first session of the re-established exchange: 905 shares of 6 listed companies [BVB press release \(2017\)](#)
- ▣ **1997**: launch of **BET** (Bucharest Exchange Trading index) on 19 Sep 1997 at 1 000 points, with 10 companies; the basket may hold 10–20 and today has 20 [BVB \(indices\)](#)
 - ▶ **BET-FI**: the first sector index, for the financial investment companies (SIFs, listed in 1999)
- ▣ **2010**: BVB lists its own shares on its **regulated market** (an EU-authorized trading venue) [BVB \(history\)](#)
- ▣ **Sep 2020**: **FTSE Russell**, a global index provider, reclassifies Romania from Frontier to (Secondary) Emerging [BVB \(history\)](#)
- ▣ **Jul 2023**: Hidroelectrica lists after the largest **IPO** (initial public offering, a first sale of shares to the public) at BVB, EUR 1.9 bn [BVB \(history\)](#)

The BET Index, 1997–2026



Reading the BET Chart

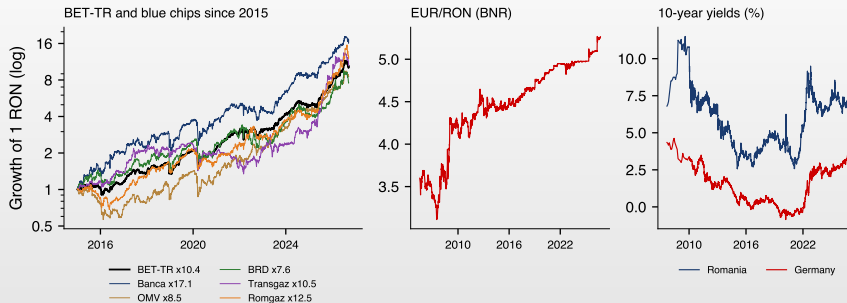
- One series: official BET closing values, 19 Sep 1997 – 18 Sep 2026; log scale, so equal distances are equal percentage changes
- Launch 1000, low 325.78 (21 Apr 1999), peak 10 813.59 (24 Jul 2007), trough 1 887.14 (25 Feb 2009): a drawdown of -82.5%
 - ▶ the 2007 peak was regained only on 16 Mar 2021, after 13.6 years
- 19 Dec 2018: -11.2% in one day (8 419.19 to 7 475.22), the day after the bank-tax announcement (Chapter 1)
- Mar 2020 (COVID-19): -31.1% from the February high to 23 Mar 2020
- 18 Sep 2026: 32 757.81, $\times 32.8$ since launch, a CAGR of 12.8% over 29 years (record 36 487.21 on 19 Aug 2026)
- Largest daily move: 7 Jan 2009, -12.3% (-13.1% log), after $+4.8\%$ the day before (official BVB daily archive BVB (indices))

 MFM_ch0_bvb_history

The Bucharest Stock Exchange

- **BVB**: Bursa de Valori București, the Bucharest Stock Exchange
- **BET**: the index of the most traded Romanian companies (blue chips); **BET-TR**: its total-return version, with reinvested dividends
- Largest companies: banks (Banca Transilvania, BRD), energy (OMV Petrom, Romgaz, Hidroelectrica), utilities (Transgaz)
- Classified as **Secondary Emerging** by [FTSE Russell \(2026\)](#), the index provider
- **TVBETETF**: an ETF traded on BVB that replicates BET, the whole index in one security

Romanian Equities, the Leu and Government Yields



Reading the Romanian Chart

- ▣ Our data: BET-TR from Sep 2014, Hidroelectrica from its listing in Jul 2023
- ▣ Growth since Jan 2015, dividend-adjusted
 - ▶ BET-TR x10.4, Banca Transilvania x17.1, Romgaz x12.5, Transgaz x10.5, OMV Petrom x8.5, BRD x7.6
- ▣ EUR/RON, official reference rate of the [BNR](#) (Banca Națională a României, the National Bank of Romania)
 - ▶ 3.60 (4 Jul 2005) to 5.26 (Sep 2026), volatility 2.1% a year since 2015
 - ▶ the new leu (RON) replaced the old leu on 1 Jul 2005 (1 RON = 10 000 ROL); the BNR archive states earlier rates in RON
- ▣ 10-year government yields, 18 Sep 2026: Romania 7.33%, Germany 3.53%
 - ▶ spread of 3.8 pp, the extra yield investors require for Romanian risk; peak 8.3 pp in Mar 2009

Data Used in This Course

Data	Content	Access in Python
Daily market data (data/market)	indices, stocks, ETFs, FX, crypto, yields	<code>pandas.read_csv</code>
FRED	US rates, macro series	public CSV
BNR	official RON reference exchange rates	public XML
DefiLlama	stablecoins, DeFi	public JSON

- Market files: one CSV per symbol, columns date, open, high, low, close, adjusted_close, volume
- All loading goes through one function, `fetch_daily(symbol, source)`
- Rule: `adjusted_close` for stocks and ETFs, `close` for indices, FX, crypto and yields
- EUR/RON always from the BNR reference rate; the market file serves only the data-pitfall example

A Minimal Python Workflow

```
import numpy as np, pandas as pd

URL = ('https://raw.githubusercontent.com/'
      'danpele/MFM/main/data/market/')

def load(sym, col='close'):
    d = pd.read_csv(URL + sym + '.csv',
                    index_col='date', parse_dates=True)
    return d[col].rename(sym)

spx = load('GSPC.INDX')           # index: close
spy = load('SPY.US', 'adjusted_close') # ETF
btc = load('BTC-USD.CC')          # 24/7
r_spx = np.log(spx).diff()        # one series
# joint analysis: join PRICES on common
# days first, then take returns
p = pd.concat([spx, btc], axis=1).dropna()
r = np.log(p).diff().dropna()
dd = spx / spx.cummax() - 1       # drawdown
```

- One series: returns on its own calendar, $\text{np.log}(s).\text{diff}()$ (the r_t of Section 2)
- Joint analysis (correlations, portfolios): join the *prices* on common days, then take returns, so both cover the same interval (Friday → Monday)
- Returns taken before the join would pair a Bitcoin Sunday → Monday move with an S&P 500 Friday → Monday move and lose Bitcoin's weekends
- $\text{spx} / \text{spx.cummax}() - 1$: the drawdown DD_t
- Crypto trades every day: annualise with 365, not 252

Data Pitfalls: Real Examples

- ▣ **Bad tick:** a wrong price in a data file, typically a spike that is reversed the next day
- ▣ EUR/RON in the market file (data/market), used only for this example
 - ▶ daily log changes of $\pm 15\%$ on 13–14 Aug 2025 and $\pm 12\%$ in early Jan 2022, each reversed the next day
 - ▶ annualised volatility 2015–2026: **12.1%** with these ticks, **2.1%** with the BNR reference rates
- ▣ Romania 10-year yield: a 0.00% print on 24 Dec 2013 and stale values on Sundays in late 2022
- ▣ A handful of bad ticks inflated the volatility estimate more than five-fold: cleaning data is part of the model

Data Checklist

- Plot the raw series and list the largest returns
- Compare two sources when possible
- Check adjustments: dividends, splits, redenominations
- Drop non-trading days and isolated outliers, and document every change
- Compute returns on each asset's own calendar before aligning assets
- Watch survivorship: delisted firms vanish from today's index lists, so past samples look too good

An Open Question: Does Passive Investing Change Price Formation?

- Passive funds buy the index in proportion to market capitalisation, whatever the information about each firm
- Haddad, Huebner & Loualiche (2025): in an estimated demand system for US institutions, active investors offset only two-thirds of a change in others' behaviour
 - ▶ the rise of passive investing over 20 years made the demand for individual stocks 11% more inelastic
- Sammon (2025): higher passive ownership makes prices anticipate earnings announcements less
- Open: the aggregate effect on concentration, volatility and price discovery
 - ▶ passive ownership is endogenous; identification relies on index reconstitutions and flow shocks

The Discovery Loop, Step by Step (1/2)

□ Literature

- ▶ List peer-reviewed papers since 2018 on passive ownership and price informativeness. For each: DOI, identification strategy, main estimate. Flag any paper you are not sure exists.
- ▶ Resolve every DOI on Crossref; the title and authors must match; read the abstract yourself

□ Hypothesis

- ▶ Give three competing explanations for SPY beating RSP after 2015 (passive flows, sector composition, earnings growth of the largest firms). For each, one testable implication the others do not share.

□ Data and code

- ▶ Write Python that reads SPY.US and RSP.US from data/market (adjusted_close), joins prices on common days, and tests H_0 : mean daily log-return spread = 0 with a Newey-West t-statistic.
- ▶ Replicate a known number first: SPY $\times 12.7$ and RSP $\times 11.9$ since April 2003 (concentration chart)

The Discovery Loop, Step by Step (2/2)

▣ Robustness

- ▶ Write a pre-registered robustness plan: subsamples, HAC bandwidths, simple vs log returns, a sector-neutral version. State in advance which outcome would falsify the hypothesis.

▣ Critique

- ▶ Act as a hostile referee. List the three strongest threats to identification in this design and the test that addresses each.

▣ Report

- ▶ Estimate, standard error, sample, every specification tried (also the failed ones)
- ▶ Prompts that shaped a decision go into AI_USE.md

- ▣ The assistant drafts; the human decides what is tested, in which order, and what counts as evidence

What the Researcher Must Check

- **Invented references:** resolve each DOI on Crossref; title, authors and journal must match
- **Wrong price field:** `close` instead of `adjusted_close` drops dividends
 - ▶ test: reproduce the total-return multiples of the concentration chart before anything else
- **Returns before the join:** mismatched calendars; test: join prices first, count the common days
- **Window shopping:** the sign of the spread depends on the start date
 - ▶ test: fix the windows in advance and report all of them
- **Low power:** “not significant” is not “no effect”
 - ▶ test: minimum detectable effect $MDE = (z_{0.975} + z_{0.80}) \times \text{s.e.} \approx 2.8 \times \text{s.e.}$ (s.e.: standard error)
- **Causality:** SPY – RSP is an outcome, not a measure of passive ownership; it needs an exogenous shifter

Mini-Case: Is There a Concentration Premium?

- A typical assistant proposal: “passive flows favour the largest firms, so cap weights beat equal weights”
- Test: $d_t = r_{SPY,t} - r_{RSP,t}$, daily log returns after joining prices on common days
 - ▶ $H_0 : \mathbb{E}[d_t] = 0$ against $H_1 : \mathbb{E}[d_t] \neq 0$; Newey–West t -statistic
- May 2003 – September 2026 (5 884 days): mean spread +0.29% a year, $t = 0.27$
- Pre-specified halves: 2003–2014 −2.28% a year ($t = -1.75$); 2015–2026 +2.85% ($t = 1.63$)
- MDE for 2015–2026: 4.9% a year, larger than the estimate itself
- Interpretation: the sign flips between halves and neither is significant
 - ▶ the “premium” is a feature of the chosen window, not evidence of a passive-investing effect

 ai_discovery_case.py

Your Turn: A Project Seed

- **Question:** does the SPY – RSP spread survive after controlling for sector returns?
 - ▶ regress d_t on the sector ETFs (XLK, XLC, XLF, ...) from data/market; test H_0 : intercept = 0
- **Replicate first:** SPY $\times 12.7$, RSP $\times 11.9$ and the mini-case t -statistics
- **Deliverables**
 - ▶ a pre-registration note (H_0 , H_1 , test, sample, corrections) committed before the results
 - ▶ code as a Quantlet; a report with estimates, standard errors and the MDE
- **Logs:** prompts that shaped decisions in AI_USE.md; every caught error (a fake reference, a wrong price field) in AI_ERRORS.md

Key Takeaways

- Markets in 2026 combine exchanges, OTC dealers, ETFs, algorithms and blockchains
- Across 2000–2026, the same patterns recur: deep drawdowns, short volatility spikes, correlations that change with the macro regime
- Every number is an estimate
 - ▶ a mean return is known to $\pm\sigma_a/\sqrt{Y}$, a Sharpe ratio to about $\pm 1/\sqrt{Y}$, whatever the sampling frequency
 - ▶ $\sqrt{h}$ scaling, correlations and drawdowns carry assumptions that the data often violate
- Passive investing increased concentration; crypto moved closer to mainstream markets via ETFs and stablecoins
- Data quality is a modelling decision: sources, adjustments and calendars change the results

Next: Chapter 1, Stylised Facts

Research questions for Chapter 1

- Is the tail index of daily returns above 4, so that the kurtosis and the autocorrelation function of r_t^2 are well defined?
- Does the autocorrelation of $|r_t|$ decay exponentially or hyperbolically (long memory)?
- At which horizon does aggregation make returns close to the Normal distribution?
- Do the same facts hold for the S&P 500, Bitcoin, BVB stocks and EUR/RON, and how do we test them with dependent, heavy-tailed data?

Further Study on Quantinar

Advanced courses on quantinar.com

-  Digital Transformation on Finance
-  Cryptos, NFTs, Digital Assets
-  Introduction to Blockchain and Cryptocurrencies
-  Statistics of Financial Markets

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